

traceAsthma: User Manual

A Mechanistic Multi-Omics Asthma Risk Model — R Package Reference

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Version 0.3.4 — July 2026

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1 Purpose

`traceAsthma` is an R package implementing **TRACE-Asthma** (Transcriptional Regulatory Activity and CpG-methylation Epigenetic Asthma score), an end-to-end computational framework that integrates genetic, epigenetic, and transcriptional regulatory information into a single, mechanistically interpretable asthma risk model. The package exists to make a specific scientific hypothesis — that inherited genetic risk for asthma is transmitted, in substantial part, through methylation-dependent transcriptional regulation rather than through fixed DNA sequence alone — reproducible, auditable, and directly usable by other researchers, rather than existing only as a private analysis script behind a single manuscript.

Concretely, `traceAsthma` provides:

- A documented, versioned implementation of every analytical step in the TRACE-Asthma pipeline, from raw quality control through a clinician-facing risk prediction interface.
- An **adaptive** entry point that works with whatever data modalities a user actually has (methylation and expression alone, with or without genotype, BMI, or smoking status), rather than requiring the full data specification.
- Realistic **simulated data**, bundled with the package, so that the entire pipeline can be exercised and verified before any real patient data is available or shared.
- Publication-standard **figures and tables**, generated directly from analysis objects, matching Nature-family journal conventions.
- A single-command **self-diagnostic** (`run_package_diagnostics()`) that verifies a fresh installation end to end.

2 Objective

The specific objectives of the package are to:

1. **Operationalize the TRACE-Asthma mechanistic architecture** —

SNP → MPRS → CpG methylation → eQTM score → TF activity → TRACE score → Asthma risk

— as a sequence of independently callable, independently testable R functions, so that each stage’s assumptions and outputs are inspectable rather than hidden inside a single monolithic analysis.

2. **Quantify, not merely assume, mediation.** Provide a structural equation modeling layer (`fit_mediation_sem()`) that formally decomposes the total effect of inherited genetic risk into a direct effect and an indirect effect transmitted through the epigenetic and transcriptional layers, with bootstrap confidence intervals.
3. **Discover, rather than presuppose, candidate regulatory mediators.** Identify candidate transcription factors from a study’s own trans-eQTM target genes via regulon enrichment testing (`test_regulon_enrichment()`), rather than selecting well-known literature TFs a priori — a design choice intended to reduce confirmation bias in the TF-discovery step.
4. **Degrade gracefully to real-world data availability.** Provide `run_trace_asthma_auto()`, which inspects whichever data modalities a user supplies and builds the best-supported model automatically, because most real-world users do not have the full genotype + methylation + expression + complete clinical panel this framework was originally specified against.
5. **Translate statistical output into actionable, plain-language information.** Provide `interpret_clinical_relevance()`, which converts a fitted model into absolute predicted

risk at defined score percentiles, a calibration trustworthiness statement, and a mechanistic mediation statement — closing the gap between “the coefficient is significant” and “here is what this means for a given patient.”

6. **Support reproducible, regulatory-quality workflows.** Provide data validation, a documented data dictionary, multi-format data import, and a realistic data simulator, so that the transition from simulated to real clinical data requires no change to analysis code — only a change in which files are loaded.

3 Novelty

As a matter of intellectual honesty, **traceAsthma**’s novelty should be understood at the level of **integration and translation**, not at the level of individual statistical methods. Every individual algorithm wrapped by this package — elastic-net regression, structural equation modeling, hypergeometric enrichment testing, VIPER-based transcription factor activity inference — is independently established and available in other R and Bioconductor packages.

The package’s genuine contributions are:

1. **A single, mechanistically ordered pipeline**, from SNP to transcriptional regulatory activity to disease risk, implemented as one coherent, testable software artifact rather than a collection of independently maintained scripts. To the authors’ knowledge, no existing R package combines multi-trait polygenic scoring, mediation-filtered eQTM discovery, data-driven transcription factor discovery, structural equation modeling of the full mediation pathway, and a deployable clinical prediction interface in one package.
2. **A data-driven, not literature-driven, transcription factor discovery step.** `test_regulon_enrichment()` nominates candidate transcription factors from a study’s own trans-eQTM target genes, which is a more defensible design for a hypothesis-generating pipeline than selecting well-known disease-associated transcription factors a priori.
3. **An adaptive pipeline architecture** (`run_trace_asthma_auto()`, `detect_available_data()`) that is, to the authors’ knowledge, unusual among domain-specific genomics pipeline packages, most of which require the full specified data set and leave graceful degradation to the user.
4. **An automated clinical-translation layer** (`interpret_clinical_relevance()`) that converts a fitted statistical model directly into a plain-language absolute-risk, calibration, and mechanistic narrative, rather than leaving that translation as a manual reporting exercise for every user.
5. **Publication output as a first-class function**, not an afterthought — Nature-journal-convention figures (600 dpi, physical sizing, colorblind-safe palettes, direct non-generic labeling) and fully footnoted tables are generated directly by the package.

Readers and reviewers should evaluate the package’s scientific claims — in particular, the magnitude of genetic-effect mediation through the eQTM/TRACE pathway — on the basis of results obtained from real cohort data, not on the basis of the software architecture described here.

4 Data Description

4.1 Bundled data (no real, proprietary, or patient-identifiable data)

`traceAsthma` includes no proprietary, confidential, or patient-identifiable data of any kind. Three data objects ship with or are generated by the package, all fully synthetic:

4.1.1 `example_cohort`

A fixed, minimal dataset (5 subjects; 2 SNPs; 2 CpGs; 4 genes; a 3-TF regulon) used for instant-run function examples and unit tests throughout the package documentation. Not sized for meaningful statistical analysis.

4.1.2 `simulated_benchmark_cohort`

A fixed-seed (20260714), realistic-scale simulated cohort bundled with the package: 300 subjects, 500 CpGs, 200 genes, 10 transcription factors, and a mild two-cluster ancestry structure. Because the seed is fixed, every user who installs the package receives an *identical* dataset, making it suitable as a shared community reference for verifying installations, comparing pipeline behavior across systems or package versions, and reporting reproducible issues.

4.1.3 `simulate_trace_asthma_cohort()`

An on-demand simulator generating a new cohort, at any user-specified size and random seed, with:

- **Genotypes** sampled under Hardy-Weinberg equilibrium with a realistic minor allele frequency spectrum (Uniform(0.05, 0.5)).
- **DNA methylation** drawn from a bimodal Beta-mixture distribution matching the marginal distribution of real Illumina EPIC/450K array data (most CpGs concentrated near 0 or 1; a minority variably methylated).
- **Gene expression** simulated as negative-binomial RNA-sequencing counts with sample-level library-size factors, matching the generative model underlying DESeq2/edgeR.
- **Clinical covariates** (age, sex, BMI, smoking status, ancestry principal components) with realistic marginal distributions and cross-correlations.
- A **known, injected causal architecture** (SNP $\rightarrow$ methylation $\rightarrow$ eQTM $\rightarrow$ TF activity $\rightarrow$ TRACE $\rightarrow$ asthma status), returned as a `ground_truth` object, enabling direct evaluation of discovery precision/recall and effect-size recovery against a known answer.

The simulator’s fidelity was verified empirically during development: at realistic sample sizes, `discover_eqtm()` run on simulated data correctly ranked injected causal CpG-gene pairs far ahead of null pairs (median $P \approx 0$ for causal pairs versus $P \approx 0.56$ for null pairs in a representative run), and `fit_risk_model()` recovered injected true effect sizes (0.40, 0.50, 0.70 for the MPRS, eQTM, and TRACE layers respectively) as approximately 0.37, 0.61, and 0.79, each correctly and strongly statistically significant.

4.2 Real-data requirements

For deployment on real cohort data, the package provides:

- `data_dictionary()` — a structured reference documenting every expected data object, field, type, unit, and coding scheme (e.g. methylation beta values in $[0, 1]$; asthma status coded

0/1; sex coded 0/1 with automatic detection of F/M text labels).

- **create_data_templates()** — generates ready-to-fill CSV templates matching the data dictionary, for preparing real data or communicating the required format to collaborators.
- **import_data()** / **export_data()** — read and write CSV, TSV, Excel, RDS, and Parquet files, and read from live database connections via DBI, so that the transition from simulated to real data requires changing only the data-loading call.
- **validate_matrix()**, **validate_clinical()**, **validate_cohort_alignment()** — structural, value-range, missingness, and cross-object subject-alignment checks, producing specific, actionable error messages (naming offending rows, columns, or subjects directly) before any downstream analysis is attempted.

No component of the analysis engine (the R/*.R source files implementing each pipeline stage) makes any reference to the specific bundled example data; every function operates purely on the shapes and types documented in the data dictionary, so real data of the correct shape is a drop-in replacement for simulated data throughout.

5 Package Methods

This section documents every analytical stage, its statistical formula, and the function implementing it. All formulas are also available with runnable worked examples in `vignette("statistical-methods")`.

5.1 Quality control

Check	Rule	Function
Variant call rate	retain if ≥ 0.95	<code>qc_genotype()</code>
Hardy-Weinberg equilibrium	retain if $P_{HWE} \geq 10^{-6}$	<code>qc_genotype()</code>
Minor allele frequency	retain if ≥ 0.01	<code>qc_genotype()</code>
Methylation probe detection	retain if fail rate ≤ 0.05	<code>qc_methylation()</code>
Gene expression	retain if count ≥ 6 in ≥ 3 samples	<code>qc_rnaseq()</code>

5.2 Multi-trait polygenic risk score (MPRS)

Trait-specific polygenic score, for individual i and trait t :

$$PRS_{i,t} = \sum_{j=1}^{m_t} \beta_{j,t} G_{ij}$$

where G_{ij} is genotype dosage and $\beta_{j,t}$ is the posterior SNP effect size for trait t (obtained upstream, e.g. via PRS-CSx/SBayesRC). Implemented in `compute_prs()`.

Multi-trait combination, via elastic-net regression against asthma status:

$$MPRS_i = \sum_t w_t PRS_{i,t}$$

Implemented in `compute_mprs()`.

5.3 Cis/trans eQTM discovery

For each candidate CpG-gene pair:

$$Y_g = \alpha_0 + \alpha_1 M_c + \mathbf{C}\gamma + \epsilon$$

where Y_g is gene expression, M_c is CpG methylation, and $\mathbf{C}$ is a covariate matrix (age, sex, ancestry principal components, cell-type composition, ...). Pairs are classified as cis if within 1 Mb of the gene's transcription start site, else trans; Benjamini-Hochberg false discovery rate control is applied separately within each class given their markedly different multiple-testing burdens. Implemented in `discover_eqtm()`.

Asthma-relevant filtering requires independent CpG- and gene-level association with asthma status:

$$\text{logit}(P(\text{Asthma})) = \beta_0 + \beta_1 \text{CpG} + \mathbf{C}\gamma$$

Implemented in `filter_asthma_relevant()`.

Mediation testing estimates the indirect effect along the pathway CpG → Expression → Asthma:

$$IE = a \times b$$

where a is the CpG-expression effect and b is the expression-asthma effect conditional on CpG, with bootstrap confidence intervals. Implemented in `test_mediation()`.

eQTM score construction applies correlation pruning ($|r| > 0.90$ collapsed to a single representative) followed by elastic-net feature selection:

$$eQTM_i = \sum_{j=1}^k w_j M_{ij}$$

Implemented in `compute_eqtm_score()`.

5.4 Regulatory network and transcription factor activity

Data-driven candidate transcription factor discovery tests each candidate TF's regulon for hypergeometric enrichment among trans-eQTM target genes:

$$P(X \geq n_{\text{overlap}}) = 1 - F_{\text{hyper}}(n_{\text{overlap}} - 1; n_{\text{target}}, n_{\text{bg}} - n_{\text{target}}, \text{regulon size})$$

Implemented in `build_regulons()` and `test_regulon_enrichment()`.

Transcription factor activity inference estimates per-subject TF activity as the normalized enrichment score of the TF's full regulon among differentially expressed genes (via VIPER or decoupleR), rather than the TF's own transcript abundance:

$$A_t = NES_t$$

Implemented in `infer_tf_activity()`. **Important implementation note:** activity inference must use each candidate TF's complete regulon, not the `target_universe`-restricted regulon used for the discovery step above; reusing the restricted regulon for inference is a documented failure mode (see package `NEWS.md`, versions 0.3.2–0.3.3) that the current implementation validates against and reports with an informative, actionable error message.

5.5 TRACE score

Constructed via a four-step procedure: univariate false-discovery-rate screening, correlation pruning, bootstrap stability selection (retaining TFs selected in a high proportion of resamples), and final elastic-net weighting on the stable TF set:

$$TRACE_i = \sum_{t=1}^m \theta_t A_{it}$$

Implemented in `compute_trace_score()`.

5.6 Mechanistic mediation (structural equation model)

$$eQTM = a_0 + a_1 MPRS + \epsilon_1$$

$$TRACE = b_0 + b_1 eQTM + \epsilon_2$$

$$Asthma = c_0 + c_1 MPRS + c_2 eQTM + c_3 TRACE + \epsilon_3$$

Direct effect = c_1 ; indirect (mediated) effect = $a_1 \times b_1 \times c_3$; total effect = direct + indirect, each with bootstrap confidence intervals. Implemented in `fit_mediation_sem()`.

5.7 Integrated asthma risk model

$$\text{logit}(P(\text{Asthma})) = \beta_0 + \beta_1 MPRS + \beta_2 eQTM + \beta_3 TRACE + \beta_4 Age + \beta_5 Sex + \beta_6 BMI + \beta_7 Smoking$$

Any subset of predictors may be supplied, accommodating sites without genotype or complete clinical data. Implemented in `fit_risk_model()`, with nested model comparison (`compare_nested_models()`) and cross-validated performance estimation (`nested_cv()`).

5.8 Validation

$$\text{AUROC} = P(\hat{p}_{\text{case}} > \hat{p}_{\text{control}})$$

$$\text{Calibration slope} = \beta_1 \text{ in } \text{logit}(\text{observed}) = \beta_0 + \beta_1 \text{logit}(\hat{p})$$

$$\text{NRI} = [P(\hat{p} \uparrow | \text{case}) - P(\hat{p} \downarrow | \text{case})] + [P(\hat{p} \downarrow | \text{control}) - P(\hat{p} \uparrow | \text{control})]$$

Implemented in `validate_model()` (AUROC, AUPRC, calibration, Hosmer-Lemeshow test, NRI, IDI) and `decision_curve()` (net benefit across risk thresholds).

5.9 Clinical interpretation

Absolute predicted risk at defined percentiles of a chosen score, holding other covariates at their sample median:

$$\Delta \hat{p} = \hat{p}(\text{score}_{90th}) - \hat{p}(\text{score}_{10th})$$

combined with a calibration trustworthiness statement and, where available, a plain-language mediation statement naming the modifiable regulatory pathway. Implemented in `interpret_clinical_relevance()`.

6 Step-by-Step Usage Guide

6.1 1. Installation

```
install.packages("remotes")
remotes::install_github("Eskezeia/traceAsthma")
library(traceAsthma)
```

6.2 2. Verify the installation

```
run_package_diagnostics()
```

Runs the full pipeline against the bundled `simulated_benchmark_cohort` and reports a PASS/FAIL/SKIPPED status per stage; stages requiring an uninstalled optional package are reported as SKIPPED with the exact install command, not as a failure.

6.3 3. Load data

```
# Bundled fixed-seed benchmark
data(simulated_benchmark_cohort)
sim <- simulated_benchmark_cohort

# Or generate a fresh simulated cohort at any size/seed
sim <- simulate_trace_asthma_cohort(n_subjects = 300, n_cpgs = 500,
                                   n_genes = 200, n_tfs = 8, seed = 42)

# Or import real data
methylation <- import_data("real_methylation.csv")
clinical <- import_data("real_clinical.csv", as_matrix = FALSE)
validate_matrix(methylation, type = "methylation")
validate_clinical(clinical)
```

6.4 4. Run the pipeline

Adaptive (recommended for most users) — automatically detects which layers your data supports:

```
result <- run_trace_asthma_auto(
  methylation = sim$methylation, expression = sim$expression,
  clinical = sim$clinical, eqtm_pairs = sim$eqtm_pairs,
  regulon_db = sim$regulon_db
)
```

Fully specified — every layer explicit:

```
result <- run_trace_asthma_pipeline(
  genotype_dosage = sim$dosage, methylation = sim$methylation,
  expression = sim$expression, eqtm_pairs = sim$eqtm_pairs,
  regulon_db = sim$regulon_db, clinical = sim$clinical,
  covariates = sim$covariates, seed = 42
)
```

6.5 5. Validate and interpret

```
pred <- predict(result$risk_model, type = "response")
val <- validate_model(result$model_data$asthma, pred)
print(interpret_clinical_relevance(result$risk_model, result$model_data, validation = val))
```

6.6 6. Generate publication output

```
export_table(format_coefficient_table(result$risk_model), "table2_coefficients.csv")
plot_trace_weights(result$trace_fit, path = "figure3_trace_weights.tiff", dpi = 600)
```

6.7 7. Deploy for prediction on new patients

```
model <- build_deployable_model(
  eqtm_weights = result$eqtm_fit$weights, tf_weights = result$trace_fit$weights,
  regulons = result$deployable_model$regulons, risk_model = result$risk_model
)
predict_asthma_risk(model, methylation = new_patient_methylation,
  expression = new_patient_expression,
  clinical = data.frame(age = 34, sex = 0))
```

6.8 Further reading

```
vignette("trace-asthma-pipeline")      # full walkthrough
vignette("statistical-methods")        # every formula, step by step
vignette("simulating-data")            # simulation and real-data transition
vignette("clinical-deployment-notes")  # validation, regulatory, publishing
```

7 Conclusion

`traceAsthma` packages the TRACE-Asthma mechanistic framework into a single, documented, versioned, and testable R package, spanning quality control, multi-layer molecular risk scoring, formal mediation modeling, prediction, validation, and publication-ready reporting. Its contribution is architectural and translational: a mechanistically ordered pipeline with a data-driven transcription factor discovery step, an adaptive interface that degrades gracefully to real-world data availability, and an automated bridge from statistical output to plain-language clinical interpretation — rather than any single novel statistical algorithm.

Every function in the package operates identically on simulated and real data, and the package ships with a realistic data simulator, a bundled fixed-seed community benchmark dataset, and a one-command self-diagnostic, so that the full pipeline can be verified end to end — by the author, by collaborators, and by any interested member of the community — before it is ever run on real patient data. Reported associations, mediated effect sizes, and predictive performance should be evaluated on the basis of results obtained from real cohort data; the software itself is offered as a transparent, reproducible instrument for obtaining and reporting those results, not as a substitute for them.

The package is under active development. Bug reports, feature requests, and contributions are welcome at <https://github.com/Eskezeia/traceAsthma>.